Structure and Reactivity of the $^1Au_6Pt$ Clusters

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In this paper we report the geometries, properties, and reactivity descriptors of 12 structural isomers located on the MP2/SDDALL potential energy surface of the $^1Au_6Pt$ binary clusters. A nonplanar, D_{3d} symmetry, cyclohexane chairlike structure is predicted to be the global minimum. Binding energies per atom in the range ≈44−51 kcal/mol account for very stable clusters. The relative stability of the clusters is directly related to all global and local reactivity descriptors. All structures are predicted to have large electron affinities. The chemical environment of the Pt atom on the structures plays a central role in the resulting relative stabilities and global and local reactivities. Our results show that more peripheral Pt atoms are more likely to be involved in electron-accepting processes.

Introduction

Since the discovery that small gold clusters and nanoparticles have catalytic activity toward CO oxidation,[1−5] several studies on the synthesis and properties of mixed metal−gold systems with improved properties have been published; the aim is to find more stable catalysts, with enhanced selectivity toward oxidation/reduction processes and with better optical and magnetic properties.[6−10] It has been shown that in contrast to bulk materials, gold clusters and nanoparticles exhibit lowly coordinated peripheral atoms whose local electronic structures increase their reactive potential;[11,12] therefore, a thorough knowledge of the potential energy surfaces of the clusters and of the electronic structures of all stable minima is necessary for the understanding of catalytic properties; such knowledge may also help design new and more efficient catalysts.

Among the variety of mixed metal−gold systems, the transition metal−gold and lanthanide−gold binary clusters are of prominent importance.[9,10,13−26] The changes in electronic structure induced by the "*impurity*" atom relative to pure gold clusters of the same total number of atoms lead to greater stabilization because of delocalization of s and d electrons on the resulting mixed clusters.[9,10,13−15] Interesting cases are the Pt−Au binary systems, which have shown potential applications in selective catalysis.[7,27−32] Specifically, the Au_6Pt system is relevant in the study of the influence of the dopant Pt atom on the geometries and electronic structures and on the modified catalytic properties as compared to those of pure Au_7 clusters.[33,34] Experimental and theoretical studies suggest that for pure Au_7 clusters, the global minimum corresponds to a C_s symmetry, edge-capped rhombus (ECR) structure.[35−39] DFT calculations show that the global minimum for the Au_7^+ PES corresponds to a planar D_{6h} structure.[40] Yuan and co-workers[33] reported the global minimum on the $^1Au_6Pt$ PES to be a D_{6h} benzene-like planar structure, with the Pt atom at the center; the calculations were performed by using the VWN and PW91 functionals for exchange and correlation, respectively, in conjunction with a triple-ζ basis set with two polarization functions, relativistic effects were accounted for within the zero-order regular approximation with no inclusion of spin−orbit coupling. On the other hand, Tian and co-workers,[34] at the BPW91/LANL2DZ level of theory found that a nonplanar, cyclohexane chairlike, D_{3d} symmetry structure to be the global minimum while the D_{6h} structure is a first-order saddle point. To our knowledge, there is no experimental nor additional theoretical evidence to support either claim; therefore, we think that more sophisticated calculations are needed to resolve this issue. Both studies, however, agree on the second most stable structure to be the edge-capped rhombus structure, ECR, with a five-coordinated Pt atom at the center.

A problematic issue in the study of molecular and atomic clusters is the generation of equilibrium structures. Recently,[41−43] a modification of the Metropolis acceptance test in the simulated annealing optimization procedure[44−46] was proposed as means for generating cluster candidate structures that undergo further optimizations by traditional gradient following techniques. The method, incorporated into the ASCEC (after its Spanish acronym Annealing Simulado Con Energía Cuántica) program,[43] retains the comparative advantages and disadvantages of stochastic optimization over analytical methods,[47] namely, initial guess independence, exhaustive exploration of the potential energy surface, and the ability to jump over energy barriers and to sample several energy wells on the same run without getting trapped in local minima; however, the method is still computationally intensive because of repetitive evaluation of the energy function. The method has been successfully applied to study the water tetramer,[41] neutral and charged lithium and bimetallic Li_5Na micro clusters,[42] the methanol tetramer,[48] the carbonic acid dimer,[49] and the water hexamer.[50] In all cases, the method affords contributing new structures never before reported in the literature, which for the appropriate cases have helped us rationalize the stabilization of hydrogen-bonded networks. In this work, we present a stochastic exploration of the $^1Au_6Pt$ potential energy surface by the ASCEC method to produce candidate structures for local minima, which are further optimized and characterized at the MP2/SDDALL level; the aim

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is to contribute to the still limited understanding of this very important PES.

Computer Methods

We used the atomic cluster capabilities of the ASCEC program, which contain an adapted version of the Simulated Annealing optimization procedure. The annealing algorithm was used to generate candidate structures after a random walk of the HF/LANL2MB $^1Au_6Pt$ PES. Further optimization and characterization of the stationary points was carried out by using second-order perturbation theory in conjunction with the Stuttgart potentials at the MP2/SDDALL level; 60 electrons were considered as core for Au and Pt atoms. Numerical harmonic second derivative calculations at the same MP2/SDDALL level were used to characterize all stationary points as true minima (no negative eigenvalues of the Hessian matrix) or saddle points. Binding energies (BE) were calculated by subtracting the energy of the cluster from the energy of the constituting atoms, in this way, larger positive numbers correspond to larger stabilization energies. Relative binding energies (ΔBE) were calculated as the difference between the energy of the most stable cluster and the energy of a particular structure. All optimization, frequency, and energy calculations were carried out using the Gaussian 03[51] suite of programs.

To obtain a comparative description of how structural issues affect the reactivity of the $^1Au_6Pt$ clusters, we followed the procedure used by Jaramillo and co-workers[52] and correlated global and local (at the platinum position) reactivity indexes with stabilization energies. Global indexes were calculated and used to compare the effect of the cluster structure and stability on its reactivity. Local indexes were used to assess the influence of the chemical environment on the reactivity of the Pt atom. We considered the indexes briefly described next. Global hardness, a measure of the chemical stability of a molecule, was calculated as[53]

$$\eta = \frac{1}{2}(\varepsilon_{\mathrm{LUMO}} - \varepsilon_{\mathrm{HOMO}}) \tag{1}$$

The Global electrophilicity index, a measure of the molecule's ability to accept electrons, was approximated by[54]

$$\omega \equiv \frac{\mu^2}{2\eta} \approx \frac{(I + A)^2}{4(I - A)} \tag{2}$$

where μ is the chemical potential, I is the ionization potential ($-\varepsilon_{\mathrm{HOMO}}$, Koopmans' theorem), and A is the electron affinity ($-\varepsilon_{\mathrm{LUMO}}$, Koopmans' theorem). The *local* electron-accepting power at the Pt atom, ω^+_{Pt}, which relates the cluster's ability to accept electron density at the location of the Pt atom, was calculated by[55]

$$\omega^+_{\mathrm{Pt}} = \omega^+ f^+_{\mathrm{Pt}} \tag{3}$$

where ω^+ is the global electron-accepting power and f^+_{Pt} is the electrophilic Fukui function condensed at the Pt atom. The condensed values at Pt sites of the acceptor Fukui function, f^+_{Pt},[56,57] were calculated by means of numerical integration over the Pt basins of the Fukui function[58] by using the Dgrid and Basin programs.[59]

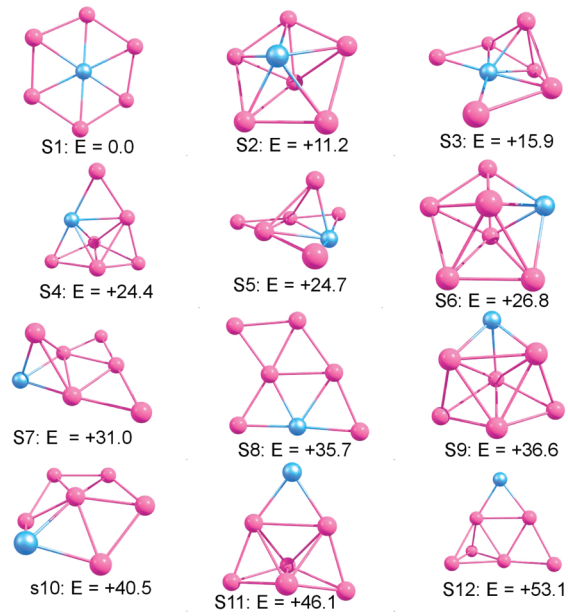

Figure 1. Local minima at the MP2/SDDALL level of theory for the $^1Au_6Pt$ potential energy surface. Notation and relative energies in kcal/mol with respect to S_1 are included. S_1 is a nonplanar, D_{3d} symmetry, cyclohexane chairlike structure with the Pt atom at the center. Lines joining atoms are drawn to help visualization of the geometrical motifs.

Results and Discussion

ASCEC Conditions. We used the big bang approach to construct the initial geometries for all ASCEC runs, namely, the seven atoms were placed at the same position, allowing them to evolve under the annealing conditions. The system was placed at the center of a cubic box of 8 Å of length; the HF/LANL2MB methodology was used to calculate the energy of a Markovian chain of randomly generated configurations; we used a geometrical quenching route with initial temperature of 400 K, a constant temperature decrease of 5% and 70 total temperatures.

Geometries. The $^1Au_6Pt$ equilibrium geometries were produced following the procedure outlined above. All geometry optimizations were carried out with no imposition of symmetry constraints as the structures coming from ASCEC are randomly generated and belong to the C_1 point group; however, some of the located stationary points have higher symmetries.

The 12 equilibrium structures located on the MP2/SDDALL potential energy surface are depicted in Figure 1. The structures were named as S_1, S_2, ..., S_{12} by their relative stability (Table 1); namely, S_1 is the most stable while S_{12} is the least stable. Only one of the geometries (S_8) is planar. Five structures (S_1, S_2, S_3, S_7, S_8) have been reported previously at different levels of theory;[33,34] three geometries (S_5, S_9, S_{10}) have been reported as being stable only in the triplet state.[34] We attempted optimizations of two additional structures reported by Tian and co-workers[34] at a lower level of theory (BPW91/LANL2DZ); the optimizations were unsuccessful because the geometries did not correspond to well-defined minima on the MP2/SDDALL PES. In all, seven structures never before reported for the $^1Au_6Pt$ clusters were located in the present work. The rich PES obtained in this study is a consequence of the stochastic nature of the search of the quantum conformational space performed by the ASCEC program, which bypasses the structure-guessing step in the search for local minima.[41,42]

The global minimum for the $^1Au_6Pt$ clusters at the MP2/SDDALL is predicted in this work to be a nonplanar, cyclo-

TABLE 1: Energetic and Geometrical Results for the $^1Au_6Pt$ Clusters at the MP2/SDDALL Level

structure	symmetry[a]	ΔE_0,[b] kcal/mol	ΔG,[c] kcal/mol	BE per atom, kcal/mol	R_{Pt-Au},[d] Å	R_{Au-Au},[d] Å	Pt coordination
S_1	D_{3d}	0.00	0.00	51.46	2.68	2.71	6
S_2	C_{5v}	11.16	12.56	49.87	2.66	2.88	6
S_3	C_s	15.91	16.32	49.19	2.64	3.00	6
S_4	C_s	24.42	25.73	47.97	2.67	2.80	5
S_5	C_s	24.74	26.31	47.93	2.68	2.89	5
S_6	C_{2v}	26.75	28.95	47.64	2.69	2.83	4
S_7	C_s	31.04	31.53	47.03	2.61	3.23	3
S^e_8	C_s	35.74	36.35	46.35	2.65	2.91	4
S_9	C_s	36.57	37.47	46.24	2.63	2.82	3
S_{10}	C_s	40.52	40.82	45.67	2.63	2.74	3
S_{11}	C_{2v}	46.14	47.67	44.87	2.55	2.79	2
S_{12}	C_s	53.13	53.73	43.87	2.54	2.75	2

[a] Closest point group. [b] Relative (ZPE corrected) energy. [c] Relative Gibbs free energy at 298.15 K, 1 atm. [d] Average pair distances. [e] S_8 is the only planar structure.

hexane chairlike, D_{3d} symmetry structure, with a 6-coordinated platinum atom at the center. A planar, benzene-like D_{6h} structure, with the 6-coordinated platinum atom at the center, corresponds to a transition state for facile interconversion between chair conformations, the energy difference between the D_{6h} saddle point and the D_{3d} minima being ≈0.4 kcal/mol. Regarding geometrical preferences of the global minimum (D_{3d} versus D_{6h}), our results are in good agreement with the report of Tian et al.[34] but are in conflict with those of Yuan and co-workers;[33] however, regarding the second most stable structure, our results are in contradiction with both reports, as the edge-capped rhombus structure, with a 5-coordinated platinum atom at the center, is predicted here to be a first-order saddle point at the MP2/SDDALL level, with an imaginary vibrational frequency of 15.0i cm^{-1}.

Table 1 lists the average distances for neighboring Pt−Au and Au−Au atom pairs. We point out that the values are comparable to the sum of atomic radii $R_{Pt} + R_{Au} = 2.70$ Å and $2R_{Au} = 2.70$ Å,[60] and to the sum of the metallic radii $R_{Pt} + R_{Au} = 2.83$ Å and $2R_{Au} = 2.88$ Å.[61] Our MP2/SDDALL calculated distances are slightly shorter than the ones previously reported with several DFT methods.[33,34]

Energies and Other Properties. The most significant energy related results are listed in Table 1. We highlighted earlier the differences and agreements of our results when compared against previous studies: regarding global minimum, a D_{3d} structure is predicted here and in Tian et al.,[34] while a D_{6h} structure is reported in Yuan et al.,[33] the second most stable structure is reported to be an edge-caped rhombus, ECR, structure in both Tian's and Yuan's works; here, we report that at the MP2/SDDALL level, a pentagonal bipyramid, PBP structure, is the second most stable, with the ECR corresponding to a first-order saddle point. A Boltzmann distribution analysis suggests that at ambient conditions (298.15 K, 1 atm), the D_{3d} global minimum would be the predominant species with populations close to 100%.

Binding energies per atom are quite large, they appear to be in the ≈44−51 kcal/mol range (Table 1). This result suggests that the $^1Au_6Pt$ bonds are relatively strong; however, small vibrational frequencies for the D_{3d} structure (lowest, 10.68 cm^{-1}, bending type; highest, 196.38 cm^{-1}, breathing mode) place some kinetic instability, which reinforces the importance of having many contributing structures to the overall stability of the cluster.

Cluster stability is directly related to the placement and coordination number of the Pt atom. A clear stabilizing trend can be seen from Table 1 and Figure 2 as the coordination number of the Pt atom increases: central (less peripheral), highly coordinated Pt atoms lead to higher stabilization energies. Planarity is not favored for the $^1Au_6Pt$ clusters at the MP2/SDDALL level, as S_8, the only located planar structure, eighth in the energy order, does not allow the Pt atom to be a 5-coordinated center; we point out that S_8 has the same geometrical motif than the most stable structure for the $^1Au_7$ cluster[36] with the Pt atom placed at the edge, not at the center. Another variable related to stabilization seems to be the average Pt−Au distance: larger distances lead to more stable structures (Table 1); this tendency is also related to the Pt coordination

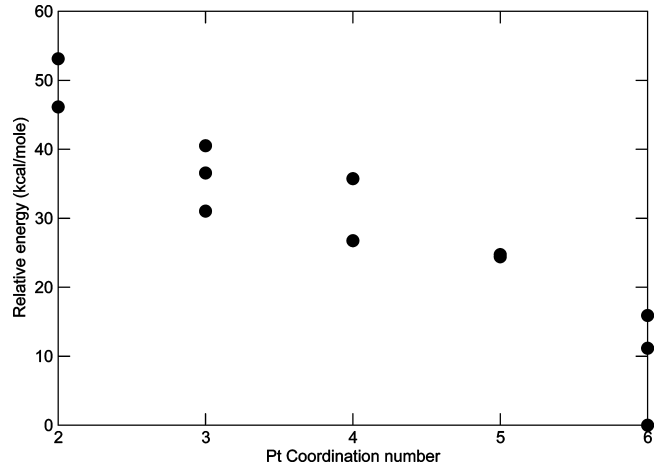

Figure 2. Relative energy as a function of the Pt coordination number for the $^1Au_6Pt$ clusters.

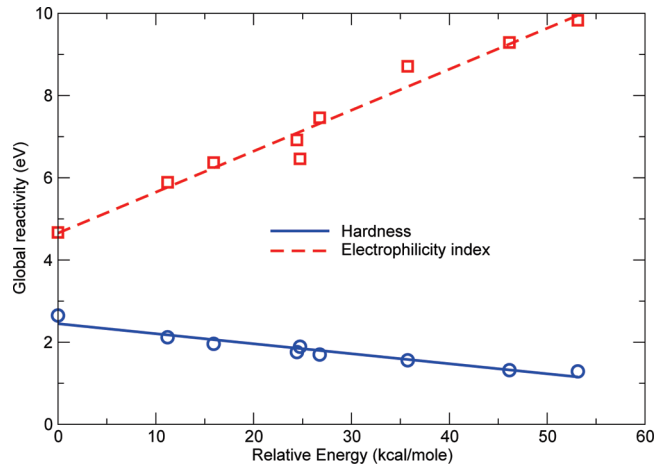

Figure 3. Global reactivity indices for the $^1Au_6Pt$ clusters. Open symbols are the calculated values; continuous and dashed lines correspond to adjusted trends. Structures S_7, S_9, and S_{10} are not considered (see text).

TABLE 2: Global Reactivity Descriptors (eV) for the $^1Au_6Pt$ Clusters

structure	μ	η	ω	ω^+
S_1	−4.97	2.65	4.67	11.06
S_2	−4.99	2.12	5.89	14.36
S_3	−5.00	1.96	6.37	15.64
S_4	−4.93	1.76	6.92	17.39
S_5	−4.94	1.89	6.46	16.08
S_6	−5.03	1.70	7.46	18.48
S_7	−5.04	2.40	5.29	12.57
S_8	−5.21	1.56	8.71	21.02
S_9	−5.04	1.90	6.67	16.31
S_{10}	−4.77	2.07	5.51	14.01
S_{11}	−4.95	1.32	9.29	23.86
S_{12}	−5.03	1.29	9.83	24.86

TABLE 3: LUMOs Energies and Local Reactivity (Electron-Accepting) at the Pt Site for Selected $^1Au_6Pt$ Clusters

structure	Pt coordination	Pt environment	ω^+_{Pt}, eV	ε_{LUMO}, eV
S_1	6	center	0.78	−0.1341
S_2	6	corner	1.17	−0.1446
S_4	5	side	1.46	−0.1490
S_7	3	corner	1.79	−0.1412

number, since a larger coordination radius is needed to accommodate a larger number of Au atoms around the Pt atom. The results in this work are by no means conclusive, further calculations including relativistic effects could improve the description of the clusters, specifically regarding the relative stability of the D_{3d} and D_{6h} structures.

Reactivity. Reactivity indexes are calculated in a very efficient and robust way in the framework of density functional theory; see, for example, the excellent review by Geerlings and others for details.[53] Accordingly, we took the MP2/SDDALL optimized geometries and calculated single point energies at the B3LYP/SDDALL level to study global and local (at the Pt site) DFT based reactivity behavior.

An important realization is that all structures have very negative LUMOs, in the range of approximately −0.13 to −0.16 eV; this provides the $^1Au_6Pt$ clusters with very high electron affinities, allowing them to easily accommodate excess electrons. We report very good linear correlations between global hardness and global electrophilicity indices with relative stabilization energies for the title systems as depicted in Figure 3 and listed in Table 2. In the present case, relative stabilities for structures S_7, S_9, S_{10}, change after B3LYP/SDDALL energies are calculated on the MP2/SDDALL optimized geometries, so we do not include the corresponding values in the plots, in an effort to avoid reoptimization of all the cluster geometries at the B3LYP level, which would produce a different PES. Figure 3 and Table 3 show that the most stable clusters are harder and have lower electrophilicity indices; that is, they are predicted to have diminished global reactivities toward electron-rich species. This observation is in excellent agreement with the above-discussed relationship between Pt coordination number and stability; that is, smaller Pt coordination numbers (more peripheral Pt, less stable structures) lead to higher global reactivity.

To analyze the effects of the chemical environment around the Pt atom on its local reactivity, we selected structures S_1 (highly coordinated, central Pt atom), S_2 (highly coordinated, Pt atom at a corner), S_4 (higly coordinated, on the side Pt atom), and S_7 (lowly coordinated, Pt atom at a corner). The ability of individual clusters to accept electron density at the Pt site is related to the *local* electron-accepting power, ω^+_{Pt}, as described above. Table 3 lists relevant information for local reactivity at the Pt site for the selected clusters. A clear trend of enhanced local reactivity for more peripheral, lowly coordinated Pt atoms emerges, which is in very good agreement with the above analyzed global reactivity issues. It is worth noticing that atom charges applied to description of local reactivity give conflicting results, an observation already pointed out by Jaramillo and co-workers[52] on the study of reactivity of substituted aromatic rings.

Local electron-accepting power, ω^+, is directly related to the placement and coordination number of the Pt atom as illustrated in Figure 4: central, less peripheral, highly coordinated Pt atoms lead to diminished electron-accepting power. We use the LUMO orbitals for the selected S_1, S_2, S_4, and S_7 structures as both global and local qualitative descriptors for reactivity toward electron-accepting processes. LUMOs for the chosen structures are plotted in Figure 5, the associated energies are listed in Table 3. The sizes of the LUMOs show an improved potential for accommodating excess electron charge around the Pt atom for the least stable (more peripheral Pt atom) structures; the calculated energies are in agreement with the global trend; that is, more stable structures lead to reduced global reactivity, except possibly for S_7. A possible explanation for this singularity was

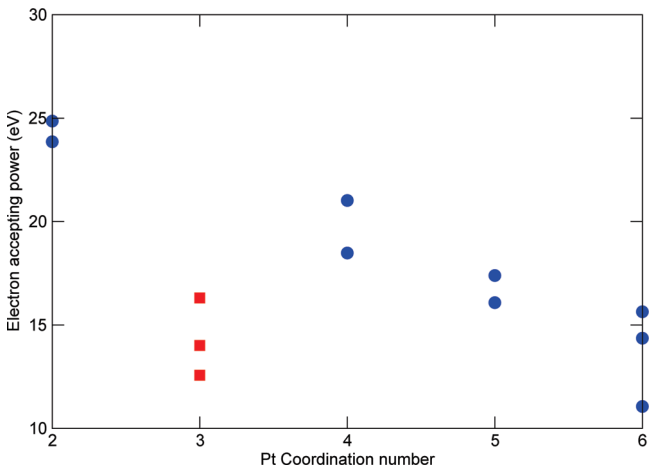

Figure 4. Electron accepting power (ω^+), as a function of the Pt coordination number for the $^1Au_6Pt$ clusters. Structures S_7, S_9, and S_{10} (red squares), change their stability order under B3LYP/SDDALL (see text).

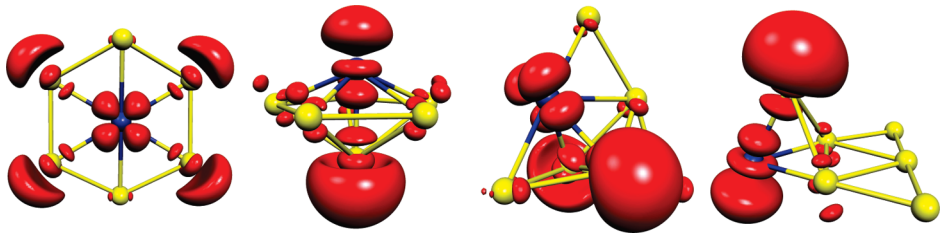

Figure 5. LUMO shapes and sizes for the S_1, S_2, S_4, and S_7 $^1Au_6Pt$ clusters, respectively. Contributions are larger from peripheral Pt atoms.

given above involving to the relative stability shuffling of the high energy structures.

Summary and Conclusions

We report the geometries, properties, and reactivity descriptors toward electron-accepting processes of 12 structural isomers located on the MP2/SDDALL potential energy surface of the $^1Au_6Pt$ binary clusters. The structures were located after random walks of the HF/LANL2MB PES subjected to a modified Metropolis acceptance test during the Simulated Annealing optimization procedure. All structures exhibit very negative LUMOs, which give them the ability to easily accommodate excess electrons. A Boltzmann distribution analysis reveals a nonplanar, D_{3d} symmetry, cyclohexane chairlike dominant structure as the global minimum. The stability order predicted in this work is in conflict with previous reports calculated at different levels of theory, except for the global minimum, which is in agreement with the results of Tian et al.[34] Binding energies per atom in the range of ≈44−51 kcal/mol account for very stable clusters. The position of the Pt atom on the structures (its chemical environment) plays a central role in the resulting relative stabilities and global and local reactivities. Relative stability of the clusters is directly related to the Pt coordination number, global hardness, global electrophilicity index, local electron-accepting power, and the energy of the LUMOs. Our results show that more peripheral Pt atoms are more likely to be involved in electron-accepting processes.

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Supporting Information Available: Cartesian coordinates for all optimized geometries reported in this paper. This material is available free of charge via the Internet at http://pubs.acs.org.

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